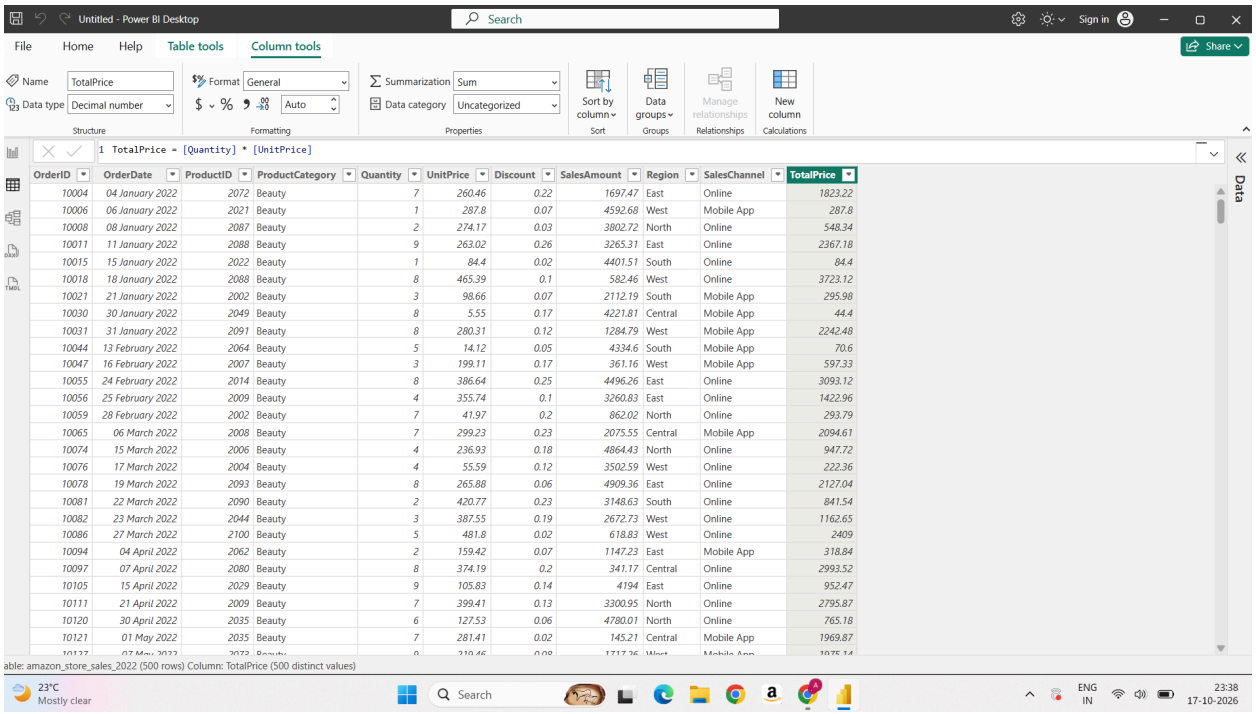
DAX Functions in Power BI

Question 1

Task: Create a calculated column TotalPrice using Quantity and UnitPrice.

ANSWER :
Code snippet

TotalPrice = [Quantity] * [UnitPrice]



OrderID	OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	TotalPrice
10004	04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	1823.22
10006	06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	287.8
10008	08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	548.34
10011	11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	2367.18
10015	15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	84.4
10018	18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	3723.12
10021	21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	295.98
10030	30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	44.4
10031	31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	2242.48
10044	13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	70.6
10047	16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	597.33
10055	24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	3093.12
10056	25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	1422.96
10059	28 February 2022	2002	Beauty	7	41.97	0.2	862.02	North	Online	293.79
10065	06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	2094.61
10074	15 March 2022	2006	Beauty	4	236.93	0.18	4864.43	North	Online	947.72
10076	17 March 2022	2004	Beauty	4	55.59	0.12	3502.59	West	Online	222.36
10078	19 March 2022	2093	Beauty	8	265.88	0.06	4909.36	East	Online	2127.04
10081	22 March 2022	2090	Beauty	2	420.77	0.23	3148.63	South	Online	841.54
10082	23 March 2022	2044	Beauty	3	387.55	0.19	2672.73	West	Online	1162.65
10086	27 March 2022	2100	Beauty	5	481.8	0.02	618.83	West	Online	2409
10094	04 April 2022	2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	318.84
10097	07 April 2022	2080	Beauty	8	374.19	0.2	341.17	Central	Online	2993.52
10105	15 April 2022	2029	Beauty	9	105.83	0.14	4194	East	Online	952.47
10111	21 April 2022	2009	Beauty	7	399.41	0.13	3300.95	North	Online	2795.87
10120	30 April 2022	2035	Beauty	6	127.53	0.06	4780.01	North	Online	765.18
10121	01 May 2022	2035	Beauty	7	281.41	0.02	145.21	Central	Mobile App	1969.87

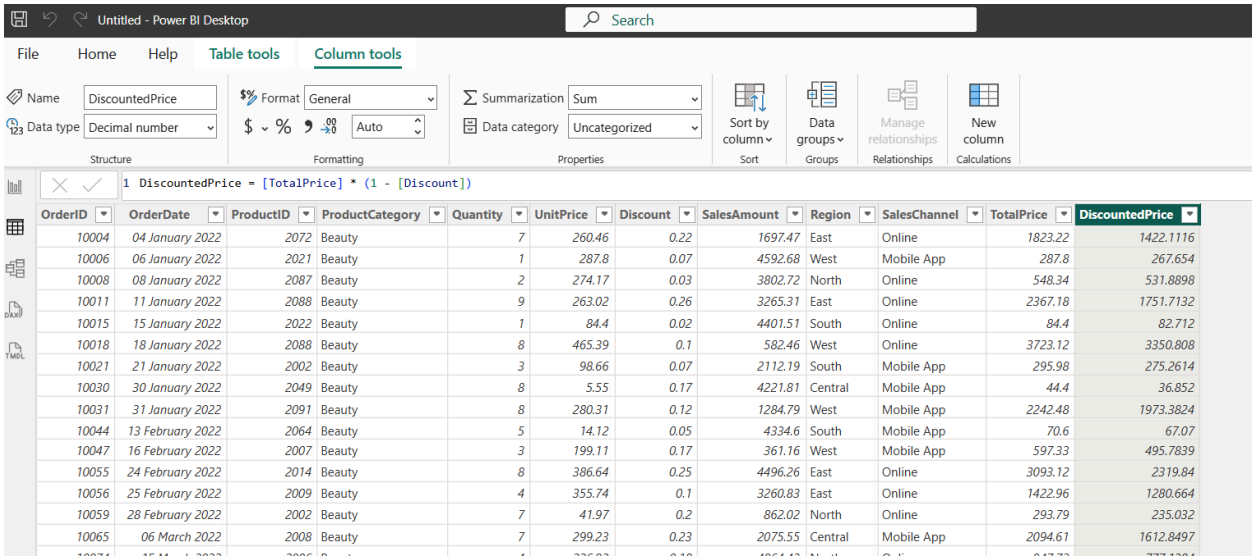
Question 2

Task: Create a calculated column DiscountedPrice.

ANSWER :

Code snippet

DiscountedPrice = [TotalPrice] * (1 - [Discount])



OrderID	OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	TotalPrice	DiscountedPrice
10004	04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	1823.22	1422.1116
10006	06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	287.8	267.654
10008	08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	548.34	531.8898
10011	11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	2367.18	1751.7132
10015	15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	84.4	82.712
10018	18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	3723.12	3350.808
10021	21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	295.98	275.2614
10030	30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	44.4	36.852
10031	31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	2242.48	1973.3824
10044	13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	70.6	67.07
10047	16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	597.33	495.7839
10055	24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	3093.12	2319.84
10056	25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	1422.96	1280.664
10059	28 February 2022	2002	Beauty	7	41.97	0.2	862.02	North	Online	293.79	235.032
10065	06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	2094.61	1612.8497

Question 3

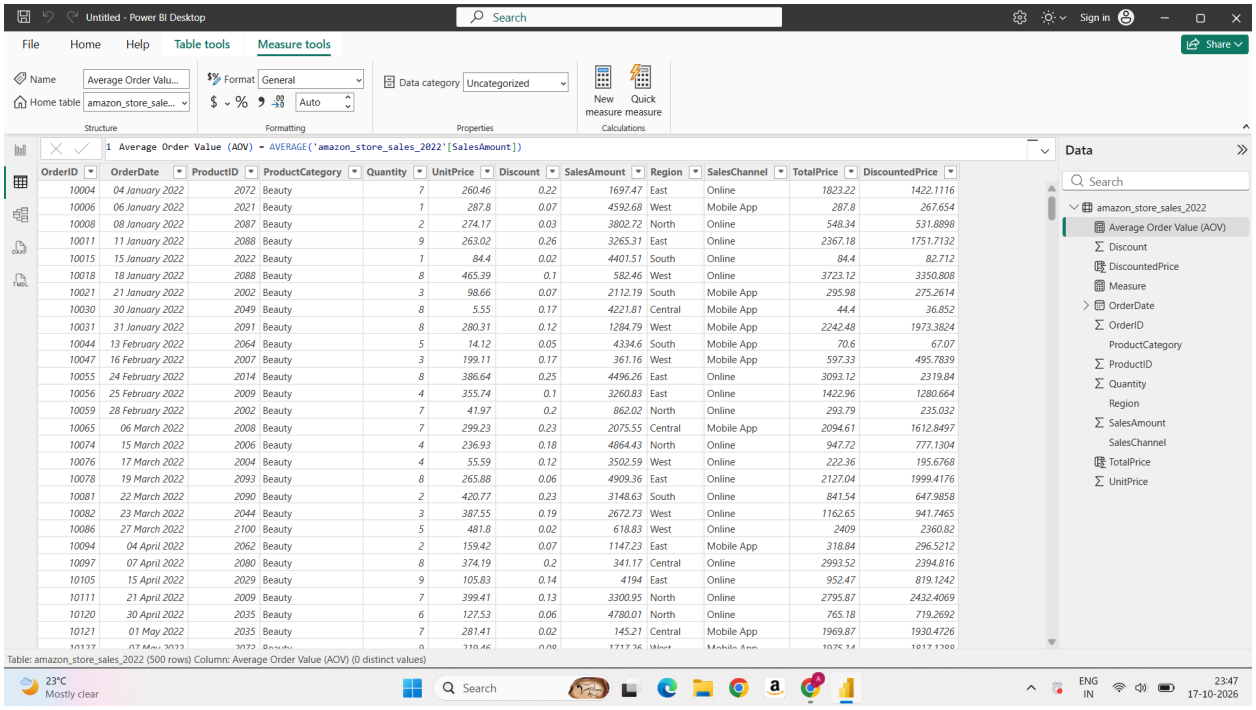
Task: Create a measure Average Order Value (AOV).

ANSWER :
Code snippet

Average Order Value (AOV) = AVERAGE(amazon_store_sales_2022[SalesAmount])

(Alternative calculation by distinct orders):

Code snippet
Average Order Value (AOV) = DIVIDE(SUM(amazon_store_sales_2022[SalesAmount]), DISTINCTCOUNT(amazon_store_sales_2022[OrderID]))



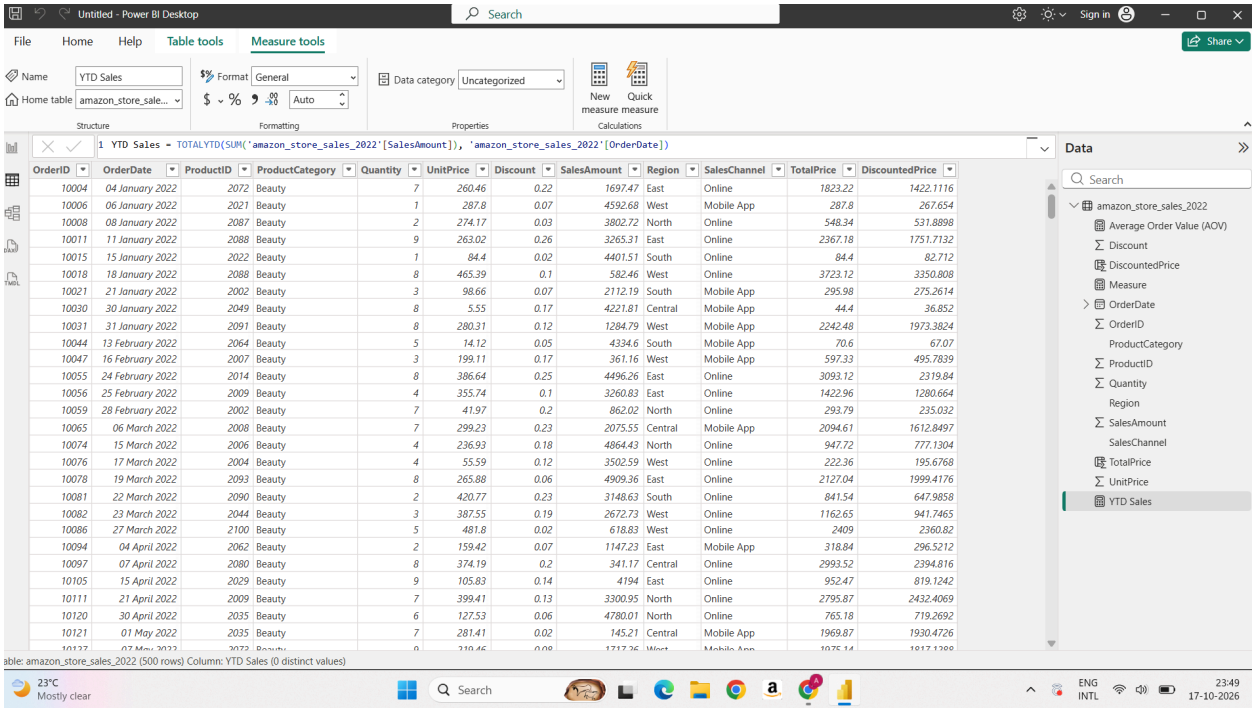
OrderID	OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	TotalPrice	DiscountedPrice
10004	04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	1823.22	1422.1116
10006	06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	287.8	267.654
10008	08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	548.34	531.8898
10011	11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	2367.18	1751.7132
10015	15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	84.4	82.712
10018	18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	3723.12	3350.808
10021	21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	295.98	275.2614
10030	30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	44.4	36.852
10031	31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	2242.48	1973.3824
10044	13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	70.6	67.07
10047	16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	597.33	495.7839
10055	24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	3093.12	2319.84
10056	25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	1422.96	1280.664
10059	28 February 2022	2002	Beauty	7	41.97	0.2	862.02	North	Online	293.79	235.032
10065	06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	2094.61	1612.8497
10074	15 March 2022	2006	Beauty	4	236.93	0.18	4864.43	North	Online	947.72	777.1304
10076	17 March 2022	2004	Beauty	4	55.59	0.12	3502.59	West	Online	222.36	195.6768
10078	19 March 2022	2093	Beauty	8	265.88	0.06	4909.36	East	Online	2127.04	1999.4176
10081	22 March 2022	2090	Beauty	2	420.77	0.23	3148.63	South	Online	841.54	647.9858
10082	23 March 2022	2044	Beauty	3	387.55	0.19	2672.73	West	Online	1162.65	941.7465
10086	27 March 2022	2100	Beauty	5	481.8	0.02	618.83	West	Online	2409	2360.82
10094	04 April 2022	2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	318.84	296.5212
10097	07 April 2022	2080	Beauty	8	374.19	0.2	341.17	Central	Online	2993.52	2394.816
10105	15 April 2022	2029	Beauty	9	105.83	0.14	4194	East	Online	952.47	819.1242
10111	21 April 2022	2009	Beauty	7	399.41	0.13	3300.95	North	Online	2795.87	2432.4069
10120	30 April 2022	2035	Beauty	6	127.53	0.06	4780.01	North	Online	765.18	719.2692
10121	01 May 2022	2035	Beauty	7	281.41	0.02	145.21	Central	Mobile App	1969.87	1930.4726

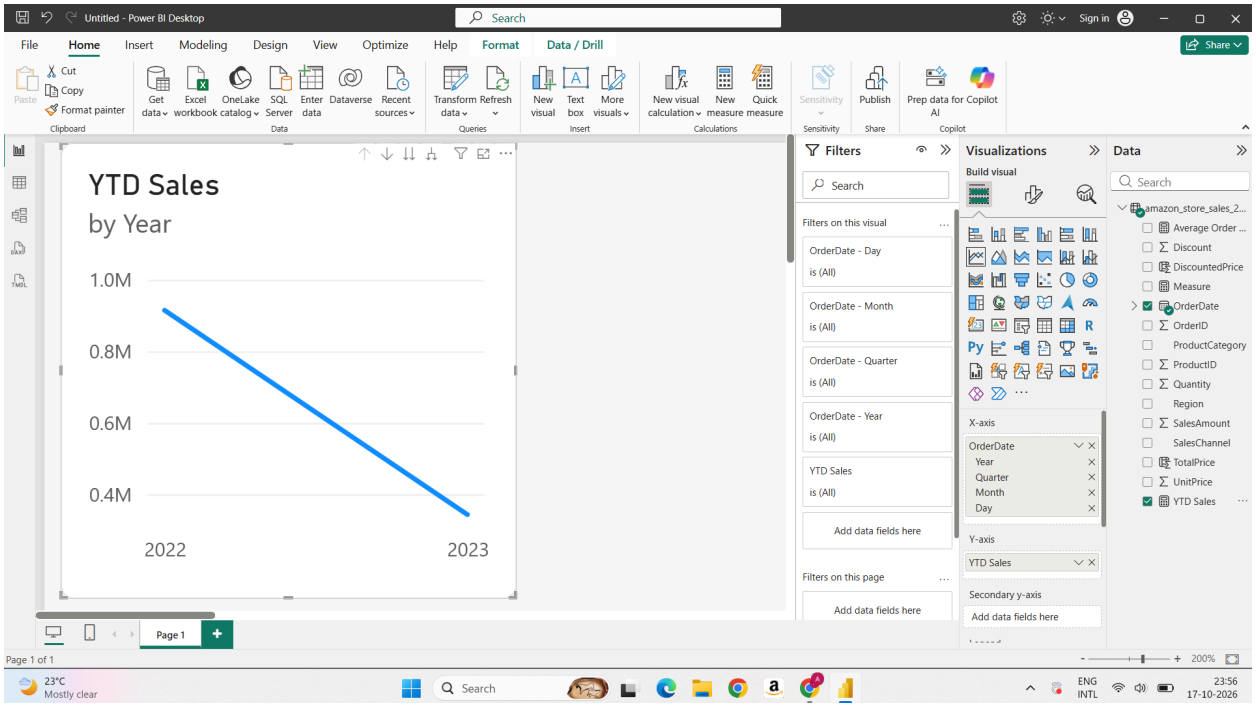
Question 4

Task: Using DAX, create a measure that calculates cumulative Year-to-Date (YTD) Sales based on the Order Date.

ANSWER :
Code snippet

YTD Sales = TOTALYTD(SUM(amazon_store_sales_2022[SalesAmount]), amazon_store_sales_2022[OrderDate])





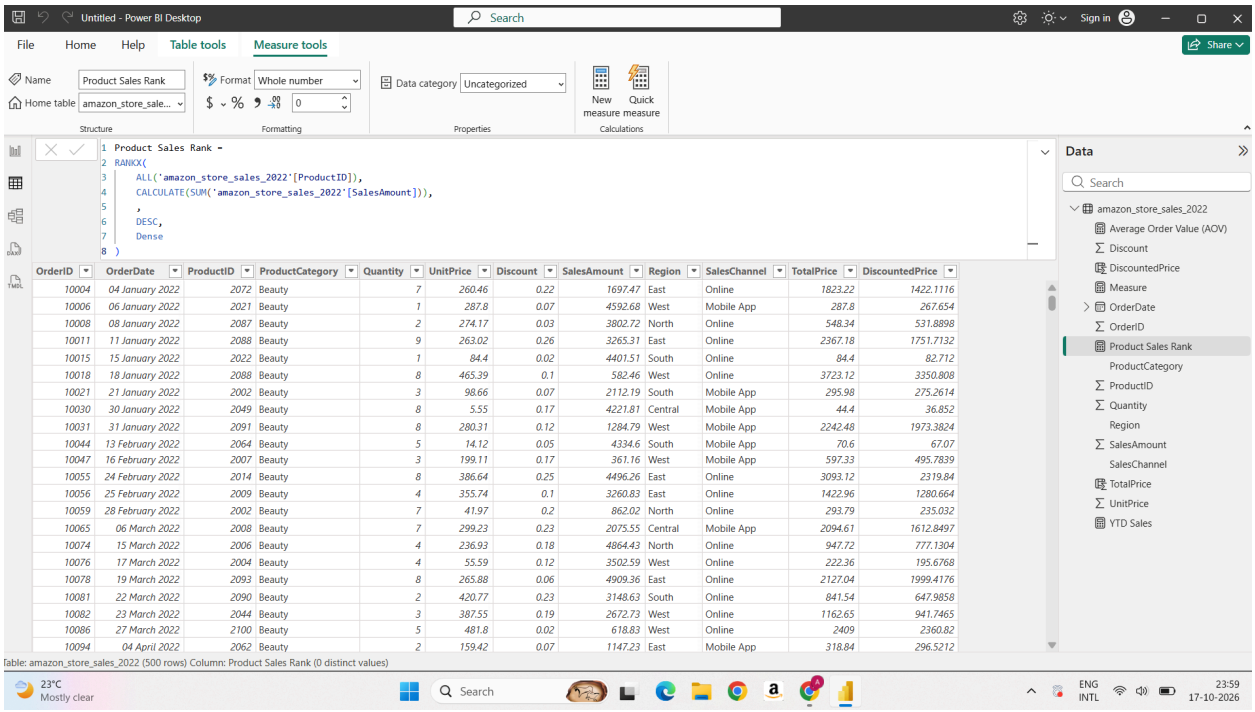
Question 5

Task: Using DAX, create a measure to rank each product according to its Total Sales, with the highest-selling product ranked first.

ANSWER :

Code snippet

```
Product Sales Rank =
RANKX(
    ALL(amazon_store_sales_2022[ProductID]),
    CALCULATE(SUM(amazon_store_sales_2022[SalesAmount])),
    ,
    DESC,
    Dense
)
```

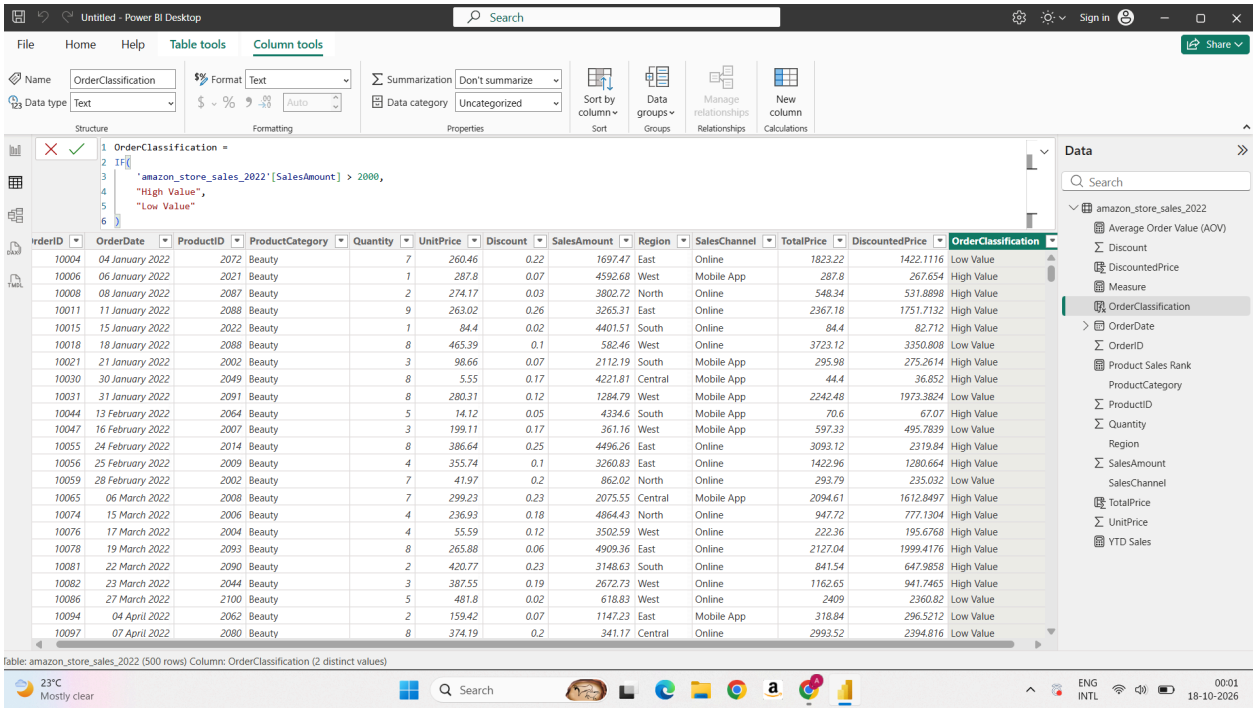


Question 6

Task: Create a calculated column classifying orders as **"High Value"** if sales > 2000, else **"Low Value"**.

ANSWER :
Code snippet

```
Order Classification =
IF(
    amazon_store_sales_2022[SalesAmount] > 2000,
    "High Value",
    "Low Value"
)
```

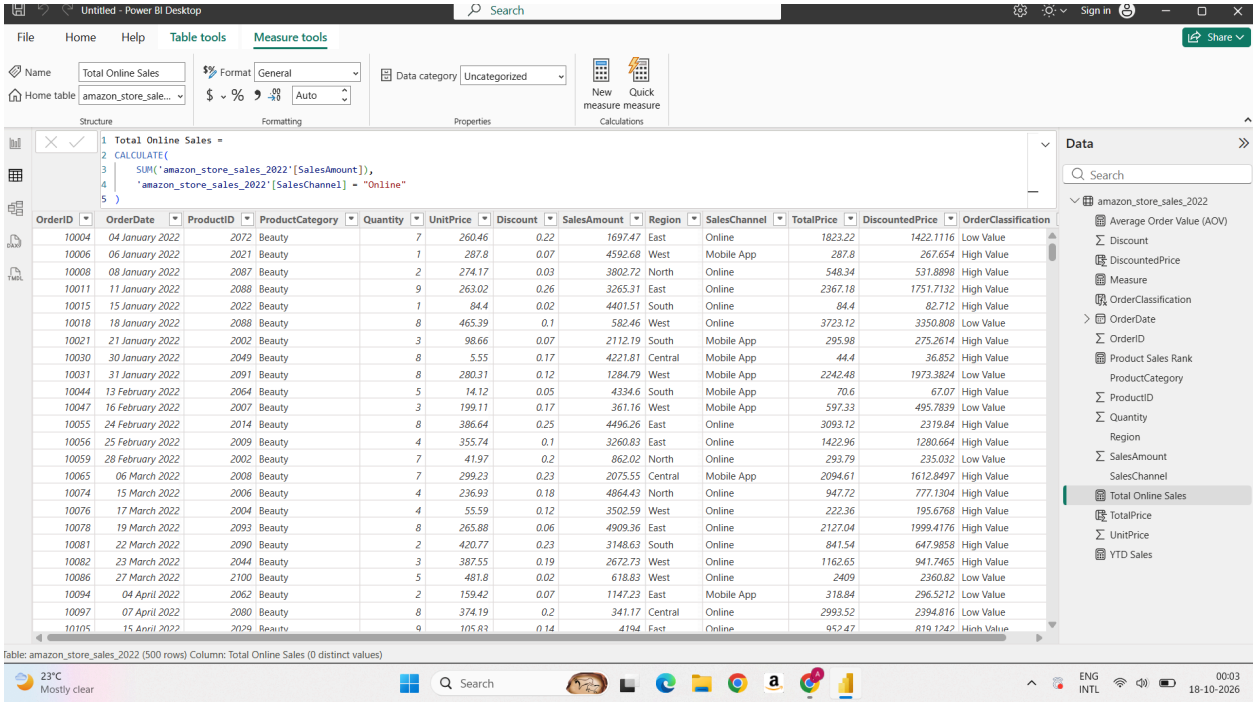


Question 7

Task: Create a measure to calculate total sales from the **Online** channel only.

ANSWER :
Code snippet

```
Total Online Sales =
CALCULATE(
    SUM(amazon_store_sales_2022[SalesAmount]),
    amazon_store_sales_2022[SalesChannel] = "Online"
)
```



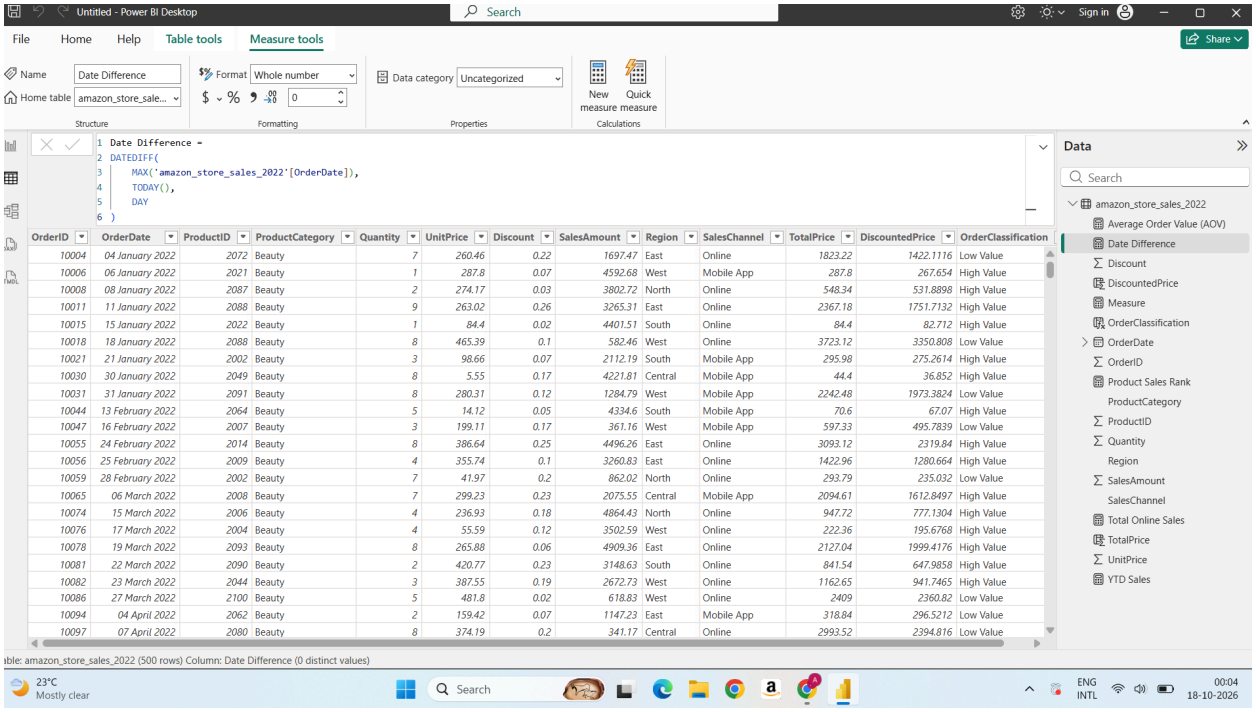
Question 8

Task: Create a measure called "Date Difference" that calculates the number of days between the latest Order Date and today's date using the DATEDIFF function.

ANSWER :

Code snippet

```
Date Difference =
DATEDIFF(
    MAX(amazon_store_sales_2022[OrderDate]),
    TODAY(),
    DAY
)
```



OrderID	OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	TotalPrice	DiscountedPrice	OrderClassification
10004	04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	1823.22	1422.1116	Low Value
10006	06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	287.8	267.654	High Value
10008	08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	548.34	531.8898	High Value
10011	11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	2367.18	1751.7132	High Value
10015	15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	84.4	82.712	High Value
10018	18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	3723.12	3350.808	Low Value
10021	21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	295.98	275.2614	High Value
10030	30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	44.4	36.852	High Value
10031	31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	2242.48	1973.3824	Low Value
10044	13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	70.6	67.07	High Value
10047	16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	597.33	495.7839	Low Value
10055	24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	3093.12	2319.84	High Value
10056	25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	1422.96	1280.664	High Value
10059	28 February 2022	2002	Beauty	7	41.97	0,2	862.02	North	Online	293.79	235.032	Low Value
10065	06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	2094.61	1612.8497	High Value
10074	15 March 2022	2006	Beauty	4	236.93	0.18	4864.43	North	Online	947.72	777.1304	High Value
10076	17 March 2022	2004	Beauty	4	55.59	0.12	3502.59	West	Online	222.36	195.6768	High Value
10078	19 March 2022	2093	Beauty	8	265.88	0.06	4909.36	East	Online	2127.04	1999.4176	High Value
10081	22 March 2022	2090	Beauty	2	420.77	0.23	3148.63	South	Online	841.54	647.9858	High Value
10082	23 March 2022	2044	Beauty	3	387.55	0.19	2672.73	West	Online	1162.65	941.7465	High Value
10086	27 March 2022	2100	Beauty	5	481.8	0.02	618.83	West	Online	2409	2360.82	Low Value
10094	04 April 2022	2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	318.84	296.5212	Low Value
10097	07 April 2022	2080	Beauty	8	374.19	0,2	341.17	Central	Online	2993.52	2394.816	Low Value